

Minimal Iterative Harnesses for Learned Reasoning: When a Tiny Transformer Becomes a Procedure Executor

Paper 0.9 — Model + Harness Computational Boundary

Matched Paper 0.85 protocol — August 2026

Abstract

Fixed Transformers have bounded forward computational depth, while many symbolic tasks have instance-dependent execution depth. We study the smallest external harness that turns a tiny Transformer from a predictor into an iterative controller, but require a five-machine comparison before attributing gains to tools: one-token output (M0), free autoregression (M1), structured scratchpad (M2), local proto-tool (M3), and explicit iterative harness (M4). The primary instrument exactly reuses Paper 0.85’s shared-vocabulary, held-out-pair implication chains. Its local R0 gate passes three seeds, yet observed model-only Stage 1 has $K_{M0} = K_{M1} = 3$ and zero recurrence gain; structured derivation supervision also stops at three. These are learned-model baselines. In contrast, the present T1/T2/T3 executions use an explicitly oracle selecting controller to validate tool semantics and cost accounting; they are not M3/M4 model results. Learned controller comparisons therefore remain pending. Tree traversal is retained as a later matched extension rather than the central initial benchmark.

1 Core idea

A fixed Transformer computes $y = F_{\theta}(x)$. The minimal harness permits

$$s_{t+1} = F_{\theta}(x, s_t),$$

where output may contain a special action token:

$$\langle CALL, op, args \rangle.$$

The harness deterministically evaluates the primitive, appends its typed result, and invokes the same model again.

The effective machine transition must be reported explicitly. M0 performs one final-token prediction. M1 emits an unconstrained proposition trace. M2 emits typed **STATE/DERIVE/ANSWER** tokens but receives no semantic callback. M3 may issue one local tool call; M4 repeatedly selects calls or an answer under an explicit step budget. A comparison that skips M1/M2 cannot distinguish external semantic execution from writable-state or supervision effects.

2 Matched propositional prerequisite and dataset

The primary experiment reuses Paper 0.85’s corrected R0-v2 and recurrence Stage1-v2. Its 64-symbol vocabulary is shared across train and test and every symbol has positive input/output training coverage. Ordered implication pairs, rather than lexical items, are disjoint: 3,226 train pairs and 806 held-out pairs with zero observed leakage. The three L2/W64/H4 R0 seeds pass entailed, consequent-swap, fact-mismatch, and rule-LHS controls at 98.8–100%. An earlier all-new-symbol condition is only a lexical/OOV-output stress and is not the local-rule prerequisite.

Proof training is bounded to $K \leq 3$. The matched initial test uses $K \in \{1, 2, 3, 4, 6, 8\}$ with held-out edges and a query instruction that does not contain the target. An earlier query-target pilot is quarantined as invalid. Future shuffled-rule, distractor, branching, and ICL cells must preserve this generator rather than substitute an easier logic task.

3 Required machine ladder and current evidence

For matched architecture and task cells define

$$K_{\text{tok}}, K_{\text{AR}}, K_{\text{scratch}}, K_{\text{tool}}, K_{\text{harness}}.$$

Table 1 distinguishes observed learned-model evidence from planned controller work.

Machine	Current condition	Measured frontier	Status
M0	O0 final-only	3	observed learned model
M1	O1 free trace	3	observed learned model
M2	O2 structure, no state labels	none	observed learned model
M2	O3 full derivation labels	3	observed learned model
M3	selected local proto-tool	—	planned accelerator
M4	iterative controller+harness	—	planned accelerator

Table 1: Evidence status for the matched ladder. Oracle-controller validation does not fill the M3/M4 rows.

M0, M1, and fully supervised M2 are perfect at $K = 1, 2, 3$ for every seed, then fail the three-seed gate at $K = 4$; the $K = 4$ acquisition fraction is zero for every condition. Hence

$$K_{\text{tok}} = K_{\text{AR}} = K_{\text{scratch}, \text{O3}} = 3, \quad G_{\text{recurrence}} = 0.$$

O1 transition accuracy falls to 23–32% at $K = 4$ and approximately 10–20% at $K = 6, 8$, with degraded termination. O2 often learns formatting/termination but no exact trajectories. This supplies a sharp model-only boundary against which M3/M4 must be compared; it does not yet show that tools expand it.

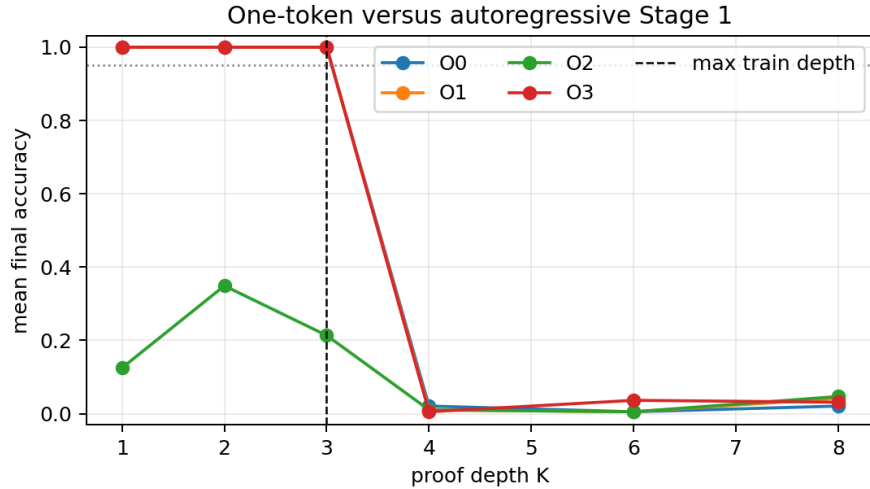


Figure 1: Observed Paper 0.85 model-only baselines reused unchanged. M3/M4 curves are intentionally absent because learned controllers have not run.

4 Minimal proto-tool interface

Use the smallest action grammar:

$$\langle \text{CALL}, op, arg_1, \dots, arg_n \rangle$$

and terminal

$$\langle \text{ANSWER}, y \rangle.$$

No planner, search algorithm, hidden chain-of-thought, retrieval system, or extra LLM is allowed in the primary tiny-model experiment.

Tool strength is a controlled resource. T1 evaluates only a controller-selected pair, $\text{apply}(\text{current}, \text{rule}) \rightarrow \text{next}$, and performs no search. T2 returns rules applicable to the current state, changing the selection/search burden. T3 returns the certified next proof state and is an upper control that moves most reasoning into the tool. The central condition begins with T1; T2 is introduced only to diagnose search, and T3 can never support the central model claim.

5 CPU-validated harness semantics

The corrected comparison infrastructure generates 384 Stage1-v2 test chains (64 at each of six depths). An explicitly oracle selecting controller executes each under T1, T2, and T3, producing 1,152 exact reference runs. All terminate correctly; calls equal K and model-invocation placeholders equal $K + 1$ by construction. These facts validate typed transitions, termination, and cost logging only. They do not estimate learned rule selection, call arguments, result integration, or stopping.

The initial learned M3/M4 slice contains 15 planned cells: five machine labels crossed with the three existing model seeds at L2/W64/H4. Model-only M0–M2 rows are imported from observed Paper 0.85 artifacts; M3/M4 remain marked `planned_accelerator`. A valid future result must separately report model forwards, generated tokens, tool calls, total context, wall time, final/transition/termination accuracy, invalid calls, and repeated-state loops.

6 Later extension: tree primitives

Begin with one primitive:

$$\text{parent}(x) \rightarrow y$$

plus a root/no-parent signal. Add children, equality, or level primitives only if necessary.

7 Later relational tasks

Compare model-only with model+harness on:

$$\text{parent}(x), \text{grandparent}(x), \text{root}(x), \text{isAncestor}(x, y),$$

$$\text{sameParent}(x, y), \text{sameGrandparent}(x, y), \text{sameAncestorAtLevel}_k(x, y).$$

Vary tree depth, branching, distractors, and required traversal independently. These experiments follow the propositional M0–M4 comparison. Existing deterministic tree reference controllers validate the harness API but are not learned-model evidence.

8 Closure hypothesis

For a fixed model, $d_{\max} < \infty$ may persist even with scale. For a harnessed model, if local transition competence is reliable,

$$T_{\text{steps}} \approx O(d)$$

should permit much larger relation depth. The key question is whether the learned controller generalizes:

call parent; update state; test stop; repeat.

9 Cross-paper benchmark families

Paper 0.5 supplies bounded pattern/refinement tasks; Paper 0.6 supplies semantic/tree relations; Paper 0.7 supplies propositional proof chains and first-order unification; Paper 0.8 supplies copy, contextual task selection, and structured rule induction. Not every task requires a tool: the objective is to determine where iteration changes the competence class.

10 Tiny-model phase diagram and pending analyses

Vary model resources L, W, H and harness budget $T_{\max}$. Compare: model-only, one callback, bounded loop, and loop until a learned stop condition.

The immediate experiment keeps L2/W64/H4 and the $1\times$ Paper 0.85 budget fixed. Only after a three-seed M3/M4 result should the design expand architecture or budget. Required analyses then separate acquisition probability from execution frontier, local rule-selection error from exact tool execution, termination from transition accuracy, observed whole-proof success from $\prod_t p_t$, and computational failure from context overflow. Mechanistic CALL/ANSWER tracing and causal patching are restricted to competent controller cells.

11 Minimality

Shrink model size, special-token grammar, tool primitives, state fields, and maximum steps. The target is the smallest model+harness pair that exhibits systematic depth extrapolation unavailable to the model alone.

12 Reproducibility and evidence boundary

Matched generator construction is in `recurrence_chains.py`. Typed T1/T2/T3 semantics are in `proof_harness.py`; the comparison and status runners use the `paper09_paper085_*` prefix. Configuration `paper085_comparison_v2.json` pins the corrected shared-vocabulary split. Artifacts under `results/paper085_comparison_v2` include the 384-instance fingerprint, 1,152 reference runs, transition/cost tables, the 15-cell plan, and a status table joining Paper 0.85 observations to pending M3/M4 work. Manifests explicitly record `oracle_controller=true` and `learned_model_results=false` for the harness validation. No tool or harness competence finding is reported in this draft.

13 Appendix: pretrained models

Use small pretrained models only as external validation, separating direct answer, proto-tool harness, native tool calling, and native reasoning/scratchpad modes where available. Corpus exposure and learned procedures remain uncontrolled.

14 Conclusion

Paper 0.9 asks where systematicity comes from when the same learned local operator is embedded in a minimal iterative machine. The present evidence fixes the baseline: local implication is reliably acquired, but ordinary one-token, free-trace, and fully supervised structured generation all stop at measured depth three. Whether a local tool or explicit loop expands that frontier, raises acquisition probability, or merely makes each selected transition exact remains an experiment rather than a conclusion. The object is not a sophisticated agent framework, but the smallest controlled resource addition that changes a measured boundary.